

Modeling Carbon Dynamics for Kettle Holes in the North-East of Germany

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Abstract

Kettle holes, that is, small ponds that formed during the final stage of the last glaciation, are prevalent in the mid and high latitudes of Europe, Asia and North America. They are considered a major sink of carbon as well as of nutrients and contaminants. Due to small size and close interconnection with the adjacent terrestrial system, they usually exhibit substantial spatial and temporal dynamics. Modeling is a way to test and improve our understanding of the related biogeochemical processes. Models should take into account interactions between multiple elements, e.g. carbon, nitrogen, phosphorus, oxygen and other chemical elements. A significant fraction of these elements originate from adjacent agricultural areas and are transported to the kettle holes via groundwater, interflow, surface runoff, wind or water erosion. This input is processed by phytoplankton, macrophytes, bacteria and fungi as the key players for biomass production and decomposition. For this, a complex eutrophication model was developed for the kettle holes surrounded by arable fields in eastern part of Germany to get an insight into the substantial short-term variations of these processes, assessing carbon budget and to better understand their effects on water quality. The model considers phytoplankton as a single group and describes its interaction with duckweed, nutrients, organic and inorganic matters in the overlying water. In addition, the model takes into consideration phosphate-iron and manganese interactions. A sensitivity analysis was conducted to determine the most sensitive parameters. This enables further exploration of the universality and robustness of the model and its parameters. Finally, the model was calibrated the observed data from an intensively studied kettle hole.

Introduction

In large areas of North Europe and North America the landscape has been shaped by several periods of glaciation (Kalettka et al., 2001; Merbach et al., 2002). During the last retreat of the ice, detached blocks of glacial ice underwent a melting which resulted in the formation of depressions that filled with water and thereafter formed what is called “kettle hole” in Europe (Kalettka and Rudat 2006; Watznauer 1989), or “potholes” in Northern America (Mitsch and Gosselink 2007). The size of the kettle holes is usually less compared to lakes and is by the German definition less than one ha, otherwise they are sorted in the lakes category (Kalettka & Rudat, 2006). It has been reported in earlier studies (Kalettka & Rudat, 2006) that there is a high number of kettle holes of between 150.000 and 300.000 with a maximum depth between 1 and 3 m located mostly in a region known by their intensive agricultural activities. These kettle holes differ widely with respect to macrophyte abundance and bank vegetation. In addition, they are rich in terms of biological diversity. Here 275 species of the macrophytes, 11 of amphibians and 16 of leeches were found (Kalettka et al., 2001; Westendorff et al., 2008). This biological diversity is mainly enhanced by the amount of nutrients from the adjacent agricultural fields which render kettle holes as hotspot of eutrophication high biomass production in the water body results in low light penetration and extreme depletion of dissolved oxygen especially near the sediment bed. Furthermore, this is accompanied with a release of phosphorus from the sediment and high suspended organic matters (Zheng-Gang, 2008). As a result, Kettle holes are considered as an ideal ecosystem for internal carbon assimilation that enhances carbon sequestration which has been substantially underestimated (Cole et al., 2007; Downing et al., 2008) and major source of greenhouse gas release. However, due to a complex interplay between internal processes and triggers from the adjacent terrestrial system kettle holes exhibit enormous spatial and temporal dynamics (Lischeid and Kalettka, 2012).

In this paper, we aim at setting up and testing a process-based model of carbon dynamics for kettle holes located in Northeast Germany. The model is based on the most prevailing processes taking place in the kettle hole (Lischeid et al., 2012) and simulates their key elements such as phytoplankton, duckweed, nitrogen, phosphorus as well as organic and inorganic matters. To the best of our knowledge this is the first model for biogeochemical processes in such a kind of system in this area. The purpose is to check whether the proposed model is able to capture most of observed dynamics and to obtain an insight into the substantial short-term vari-

ations of the processes taking place in this small system as well as their impact on long-term carbon budgets. Extensive data of an intensively studied kettle hole in this region were available for calibration and testing of the model.

Study Area

The Rittgarten kettle hole depicted in figure 1 is an eutrophic kettle hole. It is located in the catchment of the Quillow River, northwest Uckermark, about 130 km north of Berlin, Germany. The kettle hole is shallow, usually water filled with an average water depth of about 1.2 m, an area of 1497 m² and a volume of about 1797 m³ but with high water level fluctuations. It is hydraulically connected to the uppermost, possibly perched aquifer, at least during the winter season. Groundwater in this aquifer exhibits high nutrient load due to intense agriculture (Lehr et al., in preparation). It is characterised by a diversity of vegetation such as phytoplankton, periphyton, an extended reed belt and a temporary dense duckweed cover.

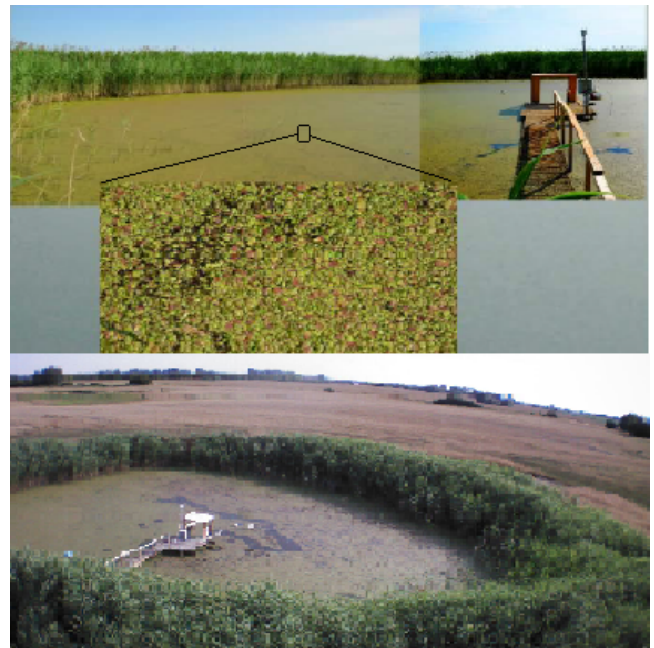


Figure 1: *Rittgarten kettle hole*

Data

Water sampling

The Rittgarten kettle hole was equipped with one YSI probe to measure light, temperature and oxygen (DO) near the center of the kettle hole. It was located ca 60 cm below the water surface. Data were recorded

every 30 minutes and a daily mean was calculated for each sampling day. In addition, a water sampling program was performed every 4 weeks from 28th February 2013 until the 10th January 2014 when three water samples were taken from the center of the kettle hole, transported to the laboratory in cooling boxes and processed within 24 hours. The water samples were filtered through $0.45\ \mu\text{m}$ cellulose nitrate filter prerinsed with $> 500\ \text{mL}$ double distilled water. The following parameters were determined from the filtrate: total dissolved organic carbon (DOC), dissolved inorganic carbon (DIC), ammonium ($\text{NH}_4 - \text{N}$), nitrate ($\text{NO}_3 - \text{N}$) and soluble reactive phosphorus (PO_4). Particulate organic carbon (POC), total nitrogen (TN) and total phosphorus (TP) were determined from unfiltered samples.

Chemical analysis of water samples: Carbon and Nitrogen

POC was determined on a precombusted *GF/C* filter after freeze-drying. DOC and TN were analyzed after acidification with a few drops of 2 M HCl until the pH was below 4 in order to remove all inorganic carbon. DIC, DOC and TN samples were then analyzed in duplicates on a Multi *N/C* 3100 Analyzer (Jena Analytics, Germany) by applying Nondispersive Infrared detection (NDIR) after combustion according to DIN EN 1484 (DEV, H3) for DIC and DOC, and according to EN 12260 (DEV, H34) for TN. $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$ were

photospectrometrically measured on a SAN++ (Skalar Analytics, The Netherlands) after EN ISO 11732 (DEV-E 23) and DIN EN ISO 13395 (DEV, D28), respectively.

Chemical analysis of water samples: Phosphorus

Water samples for TP and SRP were analyzed photospectrometrically on a CARY 1E photospectrometer (VARIAN, Germany). TP was analyzed on unfiltered samples according to EN ISO 6878 while SRP was determined on filtered samples according to DIN EN 1189 D11. The protocols for TP and SRP were slightly modified in which the molybdate solution was changed according to Murphy and Riley, (1962).

Description of the model

The model takes into account interactions between carbon, nitrogen, phosphorus, oxygen and other chemical elements. In addition, it considers phytoplankton and duckweed as organisms that assimilate carbon, nitrogen and phosphorus in the water column. The interaction of these elements are driven by forcing functions, that are, water temperature, solar radiation, water level, iron, manganese concentrations and external nutrients loading of nitrate. These interactions are illustrated in the conceptual model figure 2 and the mathematical equations are represented in the appendix A.

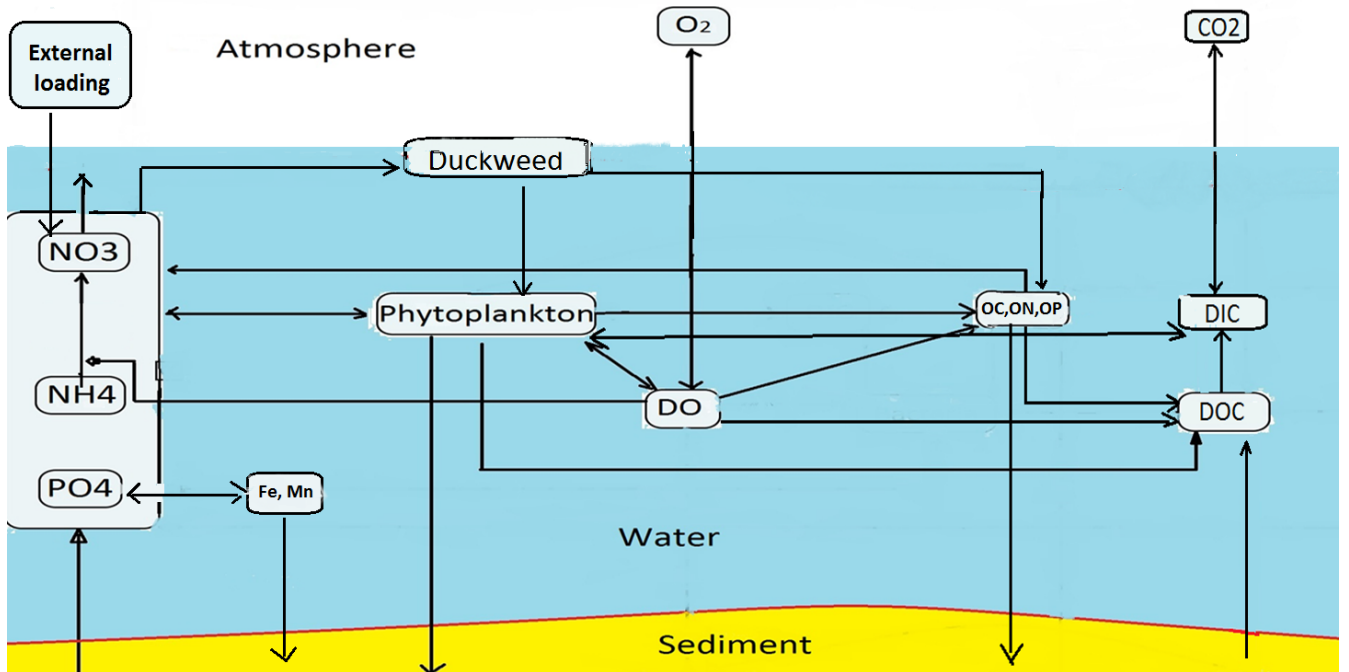


Figure 2: Relationships between organisms and chemical elements considered in the model.

Phytoplankton

For the sake of simplicity and for reducing the complexity of the model structure, the phytoplankton community is represented as one single group. The dynamic of phytoplankton biomass is described by the mass balance equation formulated by the gain and loss terms which are represented by the phytoplankton production-growth, and losses due to basal metabolism which includes excretion and respiration, mortality and the settling term for describing settling of dead phytoplankton biomass to the kettle hole sediment. The forcing factors governing the dynamic of phytoplankton biomass are solar radiation, water temperature and nutrients in the water. Light penetration is implemented using the Steele's equation (Steele, 1965). The extinction coefficient is depending on phytoplankton concentration, particulate and dissolved organic carbon and duckweed biomass. The latter affects phytoplankton biomass: The higher duckweed abundance in the surface water the lower the phytoplankton growth in the water due to shading. Water temperature is a key factor regulating the growth of phytoplankton as growth increases with temperature until an optimum is reached, after which any further temperature increase will reduce growth (Zheng-Gang, 2008). However, as the maximum water temperature in Ritgatten during the monitoring period was of about 16°C which is less than the optimal temperature, 20°C then the temperature effect is expressed only by the simple Arrhenius constant. Dissolved inorganic nitrogen and dissolved inorganic phosphorus are considered the limiting nutrients for phytoplankton growth. We also consider the dissolved inorganic carbon (DIC) to be a limiting factor for phytoplankton growth. Nitrogen and phosphorus content in the phytoplankton cells are assumed to remain constant. Here, the nutrients uptake is modelled using the Michaelis-Menten formula and Liebig's Law of the Minimum which states that phytoplankton growth is controlled by the nutrient in least supply.

Duckweed

Under suitable conditions such as sufficient availability of nutrients and temperature, the duckweed species "*Lemna minor* (L.)", have the ability to spread out over the surface of the kettle hole forming a dense cover shading deeper parts of the water column and reducing exchange of gases between atmosphere and water. In addition, duckweeds need high phosphorus and nitrogen concentrations (Portielje and Roijackers, 1995) thus competing with phytoplankton for light and nutrients uptake under the surface of the duckweeds mat. Consequently, the latters die off, leading to enhanced decomposition and oxygen consumption. The duck-

weed dynamic in this model is described by equations of growth and a loss term. Factors governing the duckweed biomass growth are water temperature, nutrients, light and a crowding term. Nutrients and crowding terms are modelled as Monod-type functions (Lüönd, 1983), while the effect of temperature is considered to be linear from the minimum temperature up to the optimum temperature (Landolt, 1986; Landolt and Kandeler, 1987). The light term is expressed by a linear function as it has been shown (Hodgson, 1970) that the photosynthesis rate increases linearly with irradiation up to a saturation value and stays constant beyond that. The loss expression includes mortality and respiration that are represented by the Arrhenius constant. In this model, we consider the duckweed as a cover that grows by uptaking nutrients (NH_4-N , NO_3-N and PO_4) from the water but we neglect considering the process by which duckweed contributes back to nutrients in the water. However, we consider its contribution to organic carbon, organic nitrogen and organic phosphorus by mortality process. (Lüönd, 1983), (Landolt, 1986; Landolt and Kandeler, 1987)

Carbon

Carbon in the water column is subdivided into four state variables: particulate organic carbon (POC), dissolved organic carbon (DOC), dissolved inorganic carbon (DIC) and total carbon (TC) which is the sum of POC, DOC and DIC. Particulate organic carbon represents the dead and alive phytoplankton carbon biomass as well as duckweed carbon biomass. Hydrolysis processes convert a fraction of particulate organic carbon to dissolved organic carbon, while another fraction settles to the sediment. The processes for dissolved organic carbon comprises phytoplankton respiration and sediment-water exchange by which DOC is released from sediment to the water as well as mineralisation to DIC. Dissolved inorganic carbon consists of carbon dioxide, bicarbonate and carbonate which is generated by photosynthesis of phytoplankton. A fraction of DIC as carbon dioxide undergoes atmosphere-water exchange using Henry's law, while another portion is released from sediment to the water column. Resuspension of particulate organic carbon from the sediment is neglected because of a dense and high reed belt that surrounds and protects the kettle hole from wind forcing.

Nitrogen

The model accounts for four nitrogen state variables that characterize the nitrogen cycle presented as organic nitrogen (ON), ammonium-nitrogen NH_4-N and nitrate-nitrogen NO_3-N and the total nitrogen (TN). Organic nitrogen is represented as a fraction of dead and

alive phytoplankton and duckweed biomass. The loss is due to decomposition to $NH_4 - N$ and sink of a particulate fraction to the sediment bed. Dissolved inorganic nitrogen as either $NH_4 - N$ or $NO_3 - N$ is assimilated by phytoplankton during growth according to the coefficient of the ammonium preference term (Thomann and Fitzpatrick, 1982; O'Connor et al., 1981; Ambrose et al., 1993) while $NH_4 - N$ is released to water by phytoplankton metabolism. $NH_4 - N$ is consumed during oxic periods by nitrification and oxidised to $NO_3 - N$, while $NO_3 - N$ is lost as nitrogen gas through denitrification during anoxic conditions. Finally, $NH_4 - N$ and $NO_3 - N$ can be released from the sediment. Resuspension of a fraction of organic nitrogen from sediment is not considered here due to the aforementioned reason.

Phosphorus

Phosphorus is considered as the primary limiting nutrient for phytoplankton and many biological elements in freshwater aquatic ecosystems (Carlson, 1977; Carpenter et al., 1998). Three state variables are considered in this model: organic phosphorus (OP), orthophosphate (PO_4) and the total phosphorus TP. Organic phosphorus is added by a fraction of dead and alive phytoplankton biomass and by duckweed phosphorus and is reduced to PO_4 by decomposition while another fraction is settling down to sediment bed. PO_4 is taken up and excreted by phytoplankton and depleted from water through adsorption onto iron and manganese oxides and hydroxides which sink to the sediment bed as phosphorus-iron and phosphorus-manganese compounds. Finally, the phosphorus cycle in the water column is enhanced by additional release of PO_4 from the sediment. Resuspension of a fraction of organic phosphorus from sediment is omitted due to the aforementioned reason.

Oxygen

The dissolved oxygen (DO) dynamic in the model depends on phytoplankton photosynthesis, respiration, atmospheric exchange of oxygen by reaeration across the air-water interface, oxygen demand for nitrification and other processes. Saturated DO concentration accounts for dependency on water temperature. Surplus oxygen degases to the atmosphere as long as not being inhibited by duckweed cover or ice. (Genet et al. (1974)).

Sediment-water exchange and external input

The interplay between sediment and the water column play a major role in shallow kettle holes. A large fraction of nutrients and organic carbon is trapped in the sediment. However, due to the limited information for porewater concentrations in Rittgarten kettle hole,

the current version of the model considers only user-specified areal nutrient fluxes to reproduce the interplay between water column and sediment. For this, release of dissolved organic and inorganic carbon, ammonium-nitrogen, nitrate-nitrogen and orthophosphate from the sediment bed were modelled as a function of temperature.

Methods

Model Implementation

Due to the large number of model parameters (ODEs), the ordinary differential equations implementation is performed using *C* programming language for the sake of simulation speed. The *R* package "deSolve" is utilized to solve the ODEs by compiling the code in *C* and loading it into *R* which allows simple interactive development and access to Rs high-level procedures (Soetaert K, Petzoldt T, Setzer R (2010a)). The model is driven by time series of forcing functions, that are, water temperature, solar radiation, water level, iron, manganese concentrations and external nutrients loading of nitrate, see figure 7. Water temperature and solar radiation time series were automatically collected on a half an hour basis by the data logger while the water level, iron and manganese data were collected on a monthly basis. External nutrients loading of nitrate-nitrogen was implemented in the model by an amount of 96 g/day from 20/03/2013 to 06/04/2013. In this study, the model is initialized on 28 February 2013 and run over one year period until 10 January 2014 using 30 minutes time step value. Model parameters were adopted from previous studies (Arhonditsis et al, 2005), (Nguyen et al, 2010), (Zeng et al, 2006), (Driever et al, 2005), (Schmitt et al, 2013) and then calibrated to fit the observed dynamics by an trial-and-error approach after the sensitivity analysis is done by selecting the most sensitive parameters. Parameters description can be found in the appendix B.

Sensitivity Analysis

Due to the large number of model parameters the most influential model parameters were identified that have the highest impact on the model state variables and their contribution to overall output uncertainty. For this purpose, a local sensitivity analysis was performed by using the function "sensFun" from R package "FME", (Soetaert and Herman 2009). Sensitivities of model output to parameters were normalised to dimensionless values as part of a sensitivity matrix whose $(i, j)^{th}$ element $SI_{i,j}$ is:

$$SI_{i,j} = \frac{\partial Y_i}{\partial X_j} * \frac{\omega_{X_j}}{\omega_{Y_i}}$$

where Y_i is an output variable, X_j is a parameter, and ω_{Y_i} is the scaling of variable Y_i which has the same physical dimension as the corresponding observation. ω_{X_j} is the scaling of parameter X_j , that is, an a priori measure of the reasonable range of X_j , in other words, the uncertainty range of parameter X_j , (Brun et al. 2001). Since it is difficult to define a specific range for all model parameters, a reasonable choice of ω_{X_j} in this study is the parameter value itself. The same applies to ω_{Y_i} , where we set it to the value of its corresponding output variable, (Brun et al. 2001).

Sensitivities $SI_{i,j}$ were calculated for all state variables and for all model parameters by allowing every parameter to vary over 10% from its nominal value as taken from the literature. Sensitivities are then ranked based on the $L2 - norm$, that is:

$$Sens_j = \sqrt{\frac{\sum_i (SI_{i,j})^2}{n}}$$

Model Calibration

Model calibration is done by trial and error approach while model performance is performed by assessing the goodness-of-fit between the simulated and monthly observed data. For this, we use two statistical methods: the root mean squared error $RMSE$ and the coefficient of determination r^2 . The former measures the mean of the squares of the errors, which is, the difference between the observed values and simulated results and it is expressed as:

$$RMSE_i = \sqrt{\frac{\sum_j (x_{(i,j)} - y_{(i,j)})^2}{n}}$$

where $RMSE_i$ is the root mean squared error value of state variable i , $x_{(i,j)}$ is the j th value of an observed result for state variable i , $y_{(i,j)}$ is the j th value of an simulated value for state variable i , and n is the number of values to be compared, that is, 13 samples per

variable. In addition, the squared Pearson correlation between simulated and observed data, that is, the coefficient of determination, is used as a common measure for model performance (Mayer and Butler, 1993), that is:

$$r^2 = \left(\frac{\text{cov}(x, y)}{\sigma_x * \sigma_y} \right)^2$$

where, cov is the covariance, σ_x and σ_y are the standard deviation of x and y respectively.

Results

Sensitivity Analysis

Figure 3 lists the five most sensitive parameters for model variables as the remaining are quite low. The Arrhenius constants for the temperature effects for some biological processes have a strong impact on all output variables without exception. In particular, the temperature-dependent coefficient for phytoplankton growth $ArrhGr$, the temperature-dependent coefficient for duckweed mortality $Arrhloss$ and duckweed growth rate $GmaxDWeed$ recorded the highest sensitivity values. In addition, The most influenced variables by these parameters are mainly duckweed, nitrate-nitrogen and ammonium-nitrogen whereas the sensitivity values for the remaining state variables are quite low. Furthermore, a clear influence of duckweed on phytoplankton can be seen by $Arrhloss$ parameter that proves the duckweed-phytoplankton interaction. These results explain the important impact of phytoplankton blooming and duckweed dynamic caused by seasonal temperature variation on the dynamics of dissolved oxygen and nutrients in the water column. Parameters related to sediment-water exchange flux of nutrients, represented by Arrhenius temperature-dependent coefficient for phosphorus flux $ArrhFluxPO4$, ammonium flux $ArrhFluxNH4$ and nitrate flux $ArrhFluxNO3$ show a significant impact on nutrients turnover.

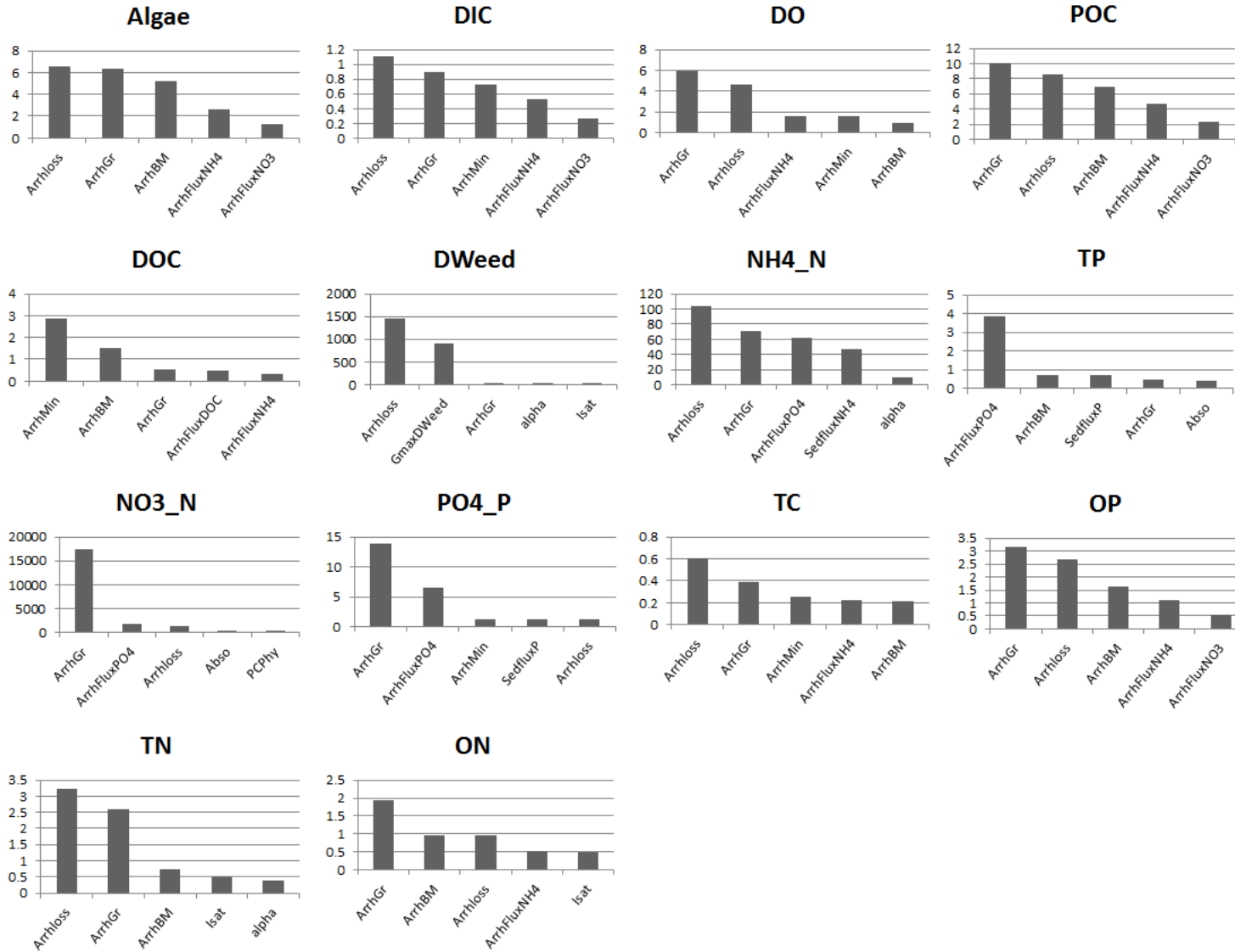


Figure 3: Sensitivity analysis results for model variables. y-axis shows sensitivity magnitudes while x-axis shows the most sensitive parameters.

Model Evaluation

Having performed the sensitivity analysis as a step to identify the most sensitive parameters that influence the model result, we now present the calibration result by comparing the simulated and observed values for Rittgarten kettle hole without external loading nitrogen input, [figure 5](#) and with external loading nitrogen input, [figure 6](#). Simulations without external loading nitrogen input did not show in general a good trends for most of the model components. Data for phytoplankton carbon biomass were not available, however simulation showed that phytoplankton carbon biomass reached $\simeq 3 \text{ mg/l}$ in early March and then declined to almost $\simeq 1 \text{ mg/l}$

in June-July before an increase occurred to $\simeq 2 \text{ mg/l}$ in September-October. For carbon biomass components, results showed that the model did not reach the particulate organic carbon peak $\simeq 11 \text{ mg/l}$ in the end of March. In addition, the model result overestimated the dissolved inorganic carbon and total carbon concentrations while underestimated the dissolved organic carbon concentration. Nitrogen and phosphorus components were slightly well estimated by the model except ammonium-nitrogen where the model underestimated its concentration.

From 20/03/2013 to 06/04/2013 an amount of 96 g/day of external loading nitrogen input was implemented to feed the model. Simulation result showed a

reasonable dynamic behaviour of phytoplankton carbon biomass described by two phases. During the first phase phytoplankton blooms took place during spring season, March-April where the concentration reached $\simeq 8 \text{ mg/l}$ and then declined due to nutrients depletion in the water column. The second phase was when the duckweed started dominating in early June, hence a decrease of phytoplankton biomass has been observed due to limited light availability. The model shows reasonable performance for most of the simulated variables. Specifically, a good agreement with the observed data was achieved for the carbon biomass components by particulate organic carbon, dissolved organic-inorganic carbon and total carbon. However, during summer period the model underestimated the simulated dissolved organic carbon values comparing to data. Particulate organic carbon was also estimated quite well for whole period of the year except in April where the model could not reach the observed high value of $\simeq 11 \text{ mg/l}$. The same applies to dissolved oxygen and organic nitrogen where the simulated concentrations could not reach the observed high oversaturation of oxygen of $\simeq 32 \text{ ml/l}$ and the high organic nitrogen concentration $\simeq 3.2 \text{ mg/l}$. In addition, simulated dissolved oxygen values were close to the observed values along the year except in September-October where anoxia was observed with $\text{DO} \simeq 0.2 \text{ mg/l}$. Simulated nitrogen components in the model showed fairly good agreement with data especially for organic nitrogen and nitrate-nitrogen. However, total nitrogen concentration was overestimated by the model in September-December by about $\simeq 1 \text{ mg/l}$. The major discrepancy in the model was observed for ammonium where the model predicted a very low concentration of ammonium-nitrogen $\simeq 0 - 0.1 \text{ mg/l}$ in the whole period of the year in contrast with the observed values that were between $\simeq 0 - 0.8 \text{ mg/l}$. Simulated phosphorus components showed a good agreement with respect to phosphate and total phosphorus except in the autumn period where they exhibited higher concentration values compared to the data. However, organic phosphorus did not show an equally good agreement with observed data throughout the year.

Figure 4 presents the model performance using the two aforementioned statistical methods. It can be seen

that the model did not perform well for the simulation without external loading nitrogen input with respect to $RMSE$ and r^2 . In contrast, significant improvement has been observed when the simulation was performed with external loading nitrogen input as the model performed reasonably well for most of state variables with respect to $RMSE$. In particular, $RMSE$ values were mainly very low for dissolved organic and inorganic carbon and total carbon. In addition, organic nitrogen, nitrate-nitrogen and total nitrogen also show low $RMSE$. In contrast, organic phosphorus, phosphate and total phosphorus exhibit high $RMSE$. However, the $RMSE$ values for particulate organic carbon and dissolved oxygen are not that low due to underprediction the high values observed in early of April. In addition, $RMSE$ for ammonium-nitrogen was fairly high because the model did not capture the highly dynamic pattern of the observed data in the summer period. The mean coefficient of determination is $r^2 = 0.78$. For some variables, the coefficient of determination confirms the $RMSE$ assessment such as inorganic carbon, total carbon, organic nitrogen, nitrate-nitrogen and total nitrogen where r^2 was greater than 0.80. The coefficient of determination was lower for dissolved oxygen and particulate organic carbon. For organic phosphorus, phosphate and total phosphorus, $RMSE$ and r^2 both gave about the same evaluation, that is, a moderate performance. The lowest r^2 was found for ammonium-nitrogen.

	Model without nitrogen loading		Model with nitrogen loading	
	r^2	$RMSE \text{ (mg/l)}$	r^2	$RMSE \text{ (mg/l)}$
DO	0.2	8.98	0.87	5.79
POC	0.01	3.35	0.87	1.5
DOC	0.03	3.26	0.69	0.89
DIC	0.59	8.16	0.85	2.52
TC	0.9	5	0.93	2.43
ON	0.05	0.69	0.88	0.3
NH4-N	0.6	0.34	0.46	0.35
NO3-N	0.99	0.03	1	0.03
TN	0.43	0.48	0.8	0.35
OP	0.16	0.08	0.6	0.05
PO4-P	0.69	0.12	0.75	0.11
TP	0.71	0.15	0.71	0.15

Figure 4: *Model performance*

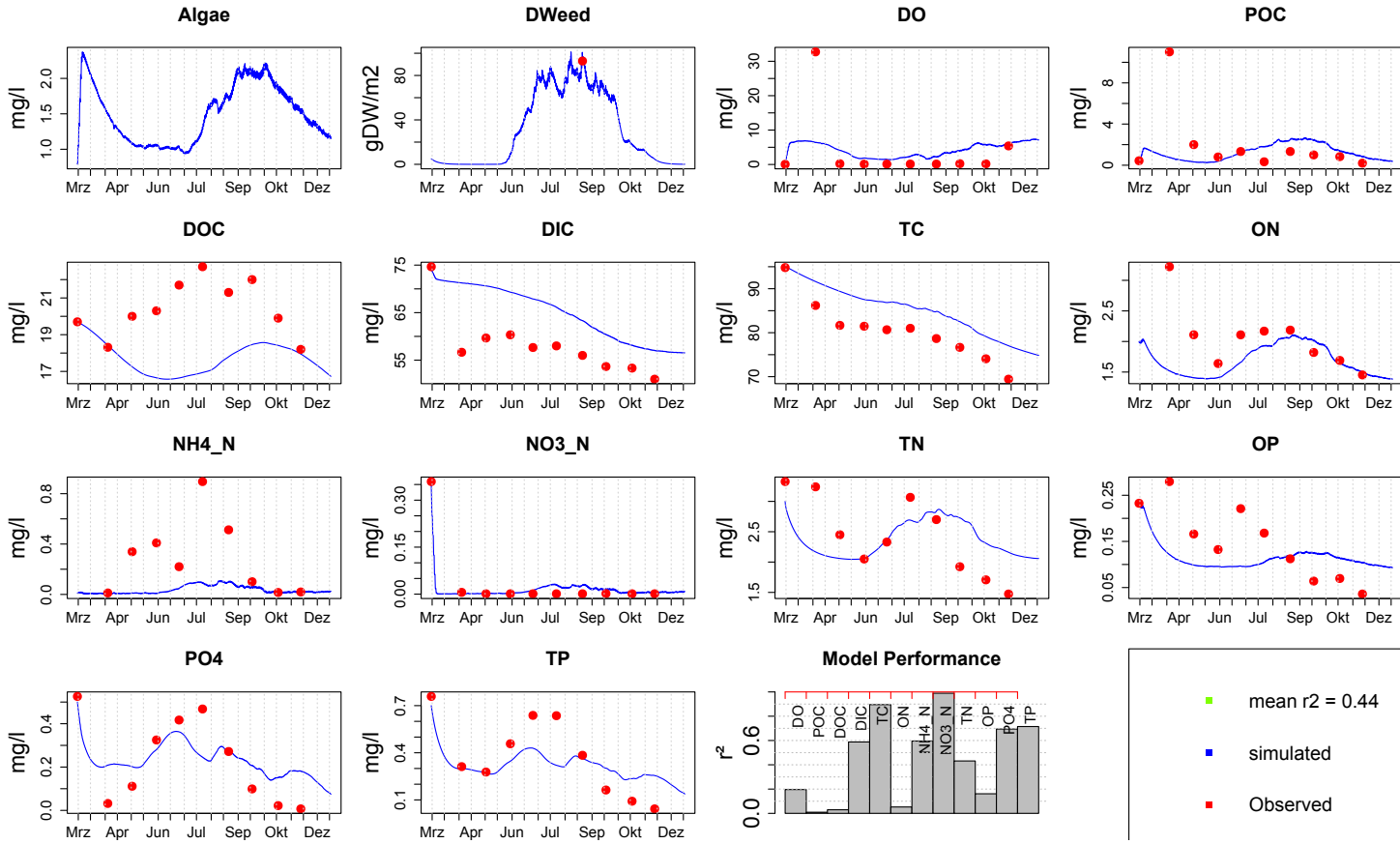


Figure 5: *Model results without external loading nitrogen input*

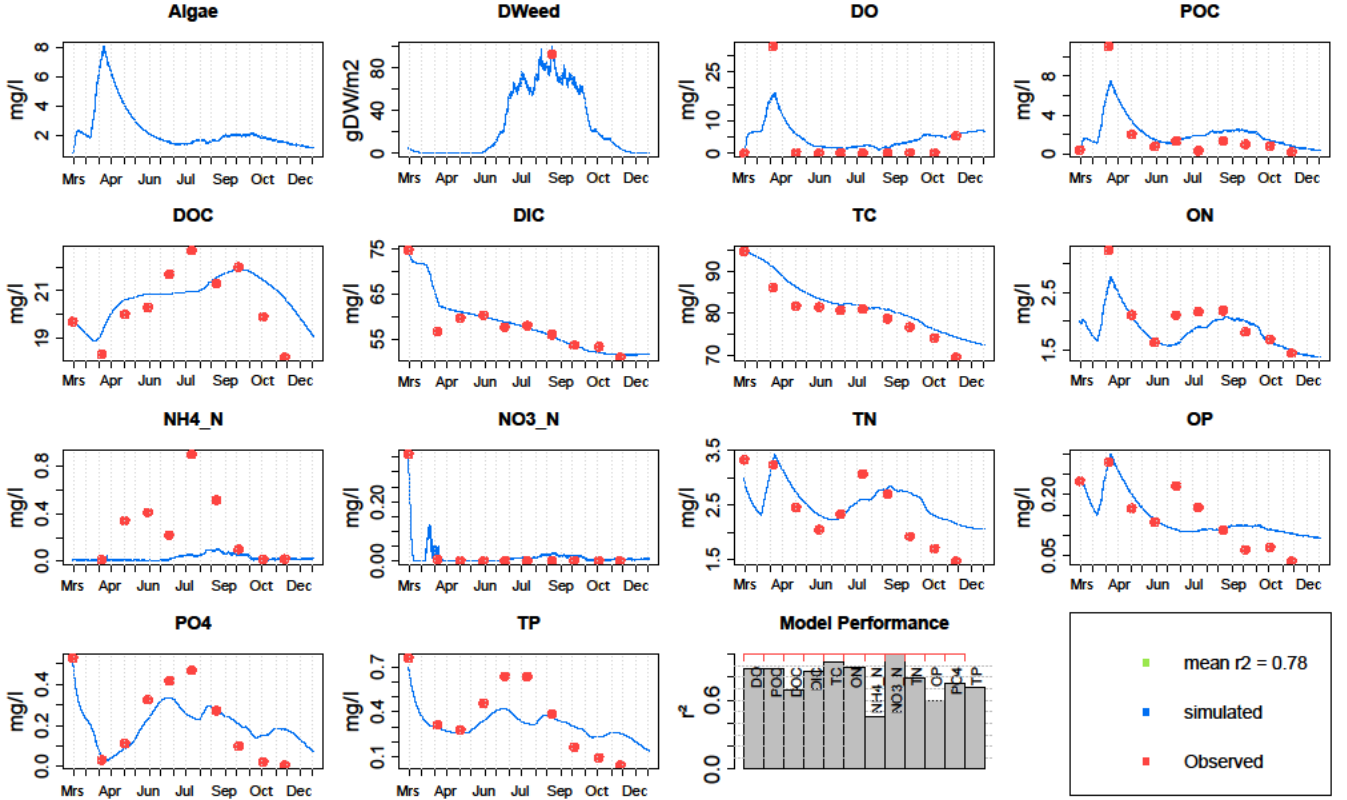


Figure 6: *Model results with external loading nitrogen input*

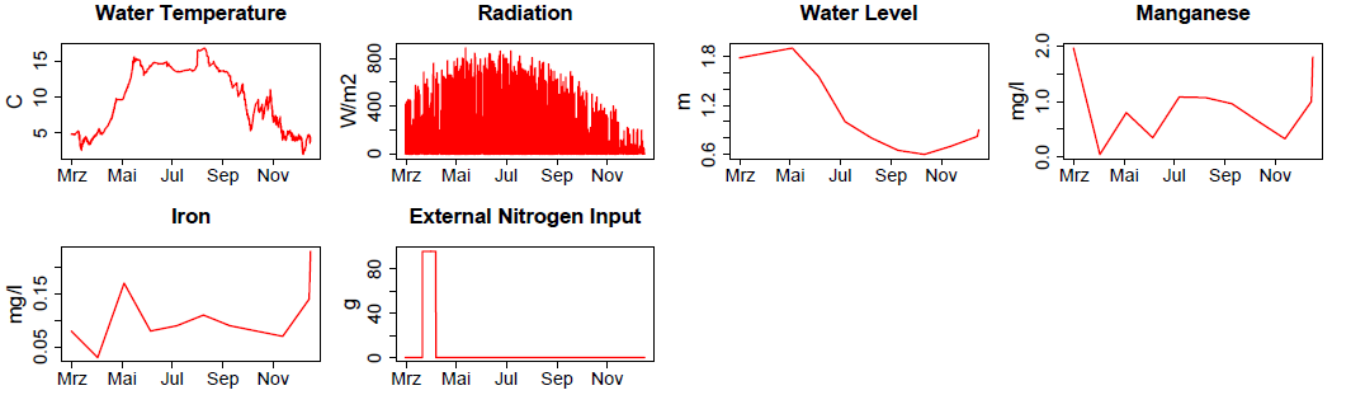


Figure 7: *Time series of the forcing functions.*

Discussion

This paper has proposed a mechanistic model that simulates phytoplankton and its interaction with duckweed as well as nutrients turnover in the studied kettle hole. Lakes have slightly different morphological and biogeochemical dynamic features comparing to our selected

system. One main characteristic is the size of the kettle hole which is much smaller than lakes with substantially more rapid dynamics and much higher sensitivity to external nutrients loading originating from agricultural surrounding area. In addition, the high water level fluctuation throughout the year is deemed to be a major environmental condition driving the model, [figure 7](#),

where the water level exhibited three different seasonal periods.

The equations of the model were derived and adopted from existing models for lakes (Arhonditsis et al, 2005), (Nguyen et al, 2010), (Zeng et al, 2006), (Driever et al, 2005), (Schmitt et al, 2013), (Lüönd, 1983), (Landolt, 1986; Landolt and Kandeler, 1987). As it has been mentioned in the model evaluation section, feeding the model with external loading nitrogen input has shown a substantial improvement in the performance of the model with the selected equations for the processes. In a duckweed dominated kettle hole, accounting for a simple equation that represents growth and loss processes showed to produce a reasonable duckweed biomass dynamic throughout the year. In addition, the formulated extinction coefficient equation for phytoplankton that accounts for duckweed, phytoplankton and organic carbon biomass resulted in a significant interaction between phytoplankton and duckweed dynamics. The mathematical equations for particulate organic carbon, nitrogen and phosphorus were expressed in terms of phytoplankton and duckweed variables. According to the model output the simulated processes seemed to produce good trends which correlate with phytoplankton and duckweed dynamics. However, regarding particulate organic carbon and nitrogen peaks depicted in the end of March and beginning of April, it is clear that the combined processes in the model could not simulate these high values which might be ascribed to the degradation of some other species such as macrophytes that are not considered in the model. The mathematical equation for phosphorus was formulated in terms of phytoplankton, sediment fluxes and iron and manganese. the combined processes were able to capture the patterns along the year. The first one was the phosphorus concentration decreases due to phytoplankton growth process in spring season. The second phenomenon was ascribed to phosphorus sediment fluxes where the water temperature reached the peak and the system was anoxic in the summer season. In fall the phytoplankton concentration was low and sediment fluxes were not strong due to the drop in water temperature, the third variation was pretty well captured and should be ascribed to iron and manganese time series that we considered in our model as a forcing functions as their concentrations were high in autumn that affected orthophosphate concentration through adsorption and settling to the bottom of sediment. Processes that were implemented for dissolved oxygen concentration exhibited a good dynamic. However, data shows that DO reached up 32 mg/l during spring season. This high value is ascribed to phytoplankton photosynthesis and respiration which was measured when phytoplankton was trapped under ice cover 4th, April 2013. It

should be noted that the model could not reach this high concentration represented by the data simply because we did not take into consideration the ice cover in our model formulation. On the other hand, the effect of duckweed on dissolved oxygen revealed in summer season where the duckweed biomass developed over this period and reached its maximum value of about $80\text{ g dry weight/m}^2$. Shading by the duckweed cover limited phytoplankton growth and decreased photosynthesis leading to oxygen depletion from the water body as the model illustrates. Nitrate-nitrogen model output showed that processes adopted in the equation were able to mimic the observed data.

During the calibration procedure, model parameters could not preserve the values drawn from the literature. Determining new parameters values by trial and error approach showed that the model output provides a reasonable results. The sensitivity analysis results support the aforementioned discussion as the most influencing parameters on dissolved oxygen were temperature-dependent coefficient for phytoplankton growth, *ArrhGr* and temperature-dependent coefficient for duckweed mortality, *Arrhloss*. In addition, results confirmed that particulate organic carbon, nitrogen and phosphorus were mainly affected by *ArrhGr*, the temperature-dependent coefficient for phytoplankton basal metabolism, *ArrhBM* and duckweed parameter, *Arrhloss*. Also, *ArrhGr* and temperature-dependent coefficient for phosphorus flux from sediment, *ArrhFluxPO4* had strong impact on the phosphorus dynamic which agrees with our previous interpretation.

The weakness of the model appeared in ammonium-nitrogen performance where the combined processes could not capture the strong fluctuation imposed by observed data that might have been affected by minor artifacts during sampling and storage. Although most of ammonium prevailing processes were included in the model, this suggests that an improvement of the mathematical structure should be adapted. In general, by borrowing some structures from mathematical models that have been exclusively designed for lakes under different environmental conditions, the model showed to produce a reasonable performance for a dominated duckweed kettle hole. However, more improvement should be considered for future developments of the model. A pillar that should be included in the model compartments is to consider a sediment submodel that could simulate the amount of organic matter and pore-water concentrations in the active layer. This will provide an evidence of the extent of sediment contribution to nutrients in the water column rather than considering user-specified areal nutrient fluxes as we adopted in the model's current version. Another important

issue is the external loadings contribution to the system. Considering some information about external loadings is inevitable issue for such a kind of small system, especially external nutrient subsidies which are mainly originated from fertilization activities by farmers. In our model, we considered a nitrate-nitrogen time serie to feed the model. As we previously stated, the model was calibrated by trial-and-error adjustment using parameters from literature. Most of these parameters however were changed after calibration. One reason we can state here is that parameters can be species-specific and site-specific which means that our model was calibrated using lakes parameters data and based on observed kettle hole data that has its specific biogeochemical dynamics behaviours. Moreover, in this study we used a local sensitivity analysis of model parameters with respect to their nominal values to just have a preliminary view about the most influencing parameters. Most of the model parameters cannot be measured directly, as they summarize effects that are not explicitly accounted for in the model.

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Appendix A: Model Equations

PHYTOPLANKTON

Temperature term

$$UTem = ArrhGr^{(temp - 20)}$$

Light term

Extinction coefficient:

$$k = k0 + k1 * CPhytoplankton + k2 * POC + k3 * DOC + k4 * DWeed$$

$$Ulight = \left(\frac{2.718}{k * H}\right) * \left(\exp\left(-\left(\frac{I0}{Iop}\right) * \exp(-k * H)\right) - \exp\left(-\frac{I0}{Iop}\right)\right)$$

Nutrient term

$$IN = NH4 + NO3$$

$$Gr = Gmax * UTem * Ulight * \min\left(\left(\frac{DIC}{DIC + Kc}\right), \left(\frac{PO4}{PO4 + Kp}\right), \left(\frac{IN}{IN + Kn}\right)\right)$$

Basal Metabolism function

$$BM = bm * ArrhBM^{(temp - 20)}$$

Settling function

$$Set = \left(\frac{Vcph}{H}\right)$$

ODE for Phytoplankton

$$dCPhytoplanktondt = Gr * CPhytoplankton - BM * CPhytoplankton - Set * CPhytoplankton$$

DuckWeed

Temperature term

$$UTemDWeed = \frac{temp - T_{min}}{T_{opt} - T_{min}}$$

Light term

$$\begin{aligned} \text{if } (I0 \leq Isat) \quad & Ligth = \alpha * I0 + \beta \\ \text{else} \quad & Ligth = 1 \end{aligned}$$

Nutrient term

$$GDWeedN = \frac{IN}{IN + K_{nDWeed}}$$
$$GDWeedP = \frac{PO4}{PO4 + K_{pDWeed}}$$

Limiting effect of biomass term

$$LimEffBiomass = \frac{KBiomass}{KBiomass + DWeed}$$

Growth functions

$$GrDWeed = G_{maxDWeed} * U_{TemDWeed} * Ligth * GDWeedN * GDWeedP * LimEffBiomass$$

Mortality functions

$$loss = loss_{weed} * Arrh_{loss}^{(temp - 20)}$$

ODE for Duckweed

$$dDWeeddt = GrDWeed * DWeed - loss * DWeed$$

Particulate organic carbon produced by died phytoplankton and duckweed:

$$POC_d$$

Hydrolysis and Settling

$$HYD = \text{hydpoc} * Tem$$

$$\text{Setpoc} = \left(\frac{V_{poc}}{H} \right)$$

ODE for POC

$$dPOC_d dt = BMPOC * BM * CPhytoplankton - HYD * POC_d - \text{Setpoc} * POC_d + \left(\frac{1}{H} \right) * CCDW * mort * loss * DWeed$$

Particulate organic carbon: died and alive phytoplankton: POC

$$dPOCdt = dCPhytoplanktondt + dPOC_d$$

Dissolved organic carbon: DOC

Mineralisation

$$Min = \text{mindoc} * TemMin * \left(\frac{DO}{DO_{satdoc} + DO} \right)$$

$$TemMin = ArrhMin^{(temp - 20)}$$

Sediment flux

$$DOCsf = \text{SedfluxDOC} * TemFlux * \left(\frac{1}{H} \right)$$

$$TemFlux = ArrhFlux^{(temp - 20)}$$

ODE for DOC

$$dDOCdt = BMDOC * BM * CPhytoplankton + HYD * POC + DOCsf - Min * DOC$$

Dissolved inorganic carbon: DIC

Surface water CO2 exchange

$$H0 = 10^{-PH}$$

$$CO2PH = \frac{H0^2}{H0^2 + (H0 * Kc1) + (Kc1 * Kc2)}$$

Sediment fluxes

$$DICsf = SedfluxDIC * TemFlux * \left(\frac{1}{H}\right)$$

ODE for DIC

$$dDICdt = (BMDIC * BM - Gr) * CPhytoplankton + Min * DOC + DICsf \\ - K * (CO2water - pCO2atm)$$

Total carbon: TC

$$dTCdt = dPOCdt + dDOCdt + dDICdt$$

Organic nitrogen produced by died phytoplankton and duckweed: ON_d

Mineralisation

$$Mindon = mindon * TemMin * \left(\frac{DO}{DOsatdon + DO}\right)$$

Settling

$$Seton = (1 - FDOM) * \left(\frac{Von}{H}\right)$$

ODE for DON

$$dON_d dt = BMDON * BM * NCPhy * CPhytoplankton - Mindon * ON_d - Seton * ON_d + \\ \left(\frac{1}{H}\right) * NCDW * mort * loss * DWeed$$

Organic nitrogen: died and alive phytoplankton: ON

$$dONdt = dON_d + NCPhy * dCPhytoplanktondt$$

Ammonium-Nitrogen: NH₄_N

Nitrification

$$TemNit = ArrhNitr^{(temp - 20)}$$

$$Nitr = Nitrmax * \left(\frac{DO}{KhnitDo + DO} \right) * TemNit$$

Sediment flux

$$NH4sf = SedfluxNH4 * TemFlux * \left(\frac{1}{H} \right)$$

Nitrogen preference

$$r = \frac{NH4 * NO3}{(NH4 + Kn) * (NO3 + Kn)} + \frac{NH4 * Kn}{(NH4 + NO3) * (NO3 + Kn)}$$

ODE for NH₄_N

$$\begin{aligned} dNH4dt = & (BMNH4 * BM - r * Gr) * NCPHy * CPhytoplankton + \\ & Mindon * ON - Nitr * NH4 + NH4sf \end{aligned}$$

Nitrate-Nitrogen: NO₃_N

Sediment fluxes

$$NO3sf = SedfluxNO3 * TemFlux * \left(\frac{1}{H} \right)$$

Denitrification

$$Denit = denit * ArrhDent^{(temp-20)} * \left(\frac{DOdent}{DOdent+DO} \right)$$

ODE for NO₃

$$\begin{aligned} dNO3dt = & - (1 - r) * Gr * NCPHy * CPhytoplankton \\ & + Nitr * NH4 - Denitr * NO3 + NO3sf \end{aligned}$$

Total Nitrogen: TN

$$dTNDt = dONdt + dNH_4dt + dNO_3dt$$

Organic Phosphorus produced by died phytoplankton: OP_d

Mineralisation

$$Mindop = mindop * TemMin * \left(\frac{DO}{DO_{satdop} + DO} \right)$$

Settling

$$Seton = (1 - FDOM) * \left(\frac{Vop}{H} \right)$$

ODE for OP

$$dOP_d dt = BMDOP * BM * PCPhy * CPhytoplankton - Mindop * OP_d + setpop * OP_d + \left(\frac{1}{H} \right) * PCDW * mort * loss * DWeed$$

Organic Phosphorus: died and alive phytoplankton: OP

$$dOPdt = dOP_d + PCPhy * CPhytoplankton$$

Phosphate_Phosphorus: PO_4_P

Sediment fluxes

$$Psf = SedfluxP * TemFlux * \left(\frac{1}{H}\right)$$

ODE for PO4_P

$$dPO4dt = (BMP04 * BM - Gr) * PCPhy * CPhytoplankton + \\ Mindop * OP + Psf - \left(\frac{Abso}{H}\right) * Vset * Fe * PO4$$

Total Phosphorus: TP

$$dTPdt = dOPdt + dPO4dt$$

Dissolved Oxygen: DO

DO consumption in sediment

$$DOSed = SOD * \frac{ArrhDO^{(temp-20)}}{H}$$

DO saturation

$$DOsat = -0.000063 * temp^3 + 0.0054258 * temp^2 - 0.68217 * temp + 10.5532$$

ODE for DO

$$dDOdt = Rea * (DOsat - DO) + (Gr - BMDIC * BM) * ResDOC * CPhytoplankton \\ - Min * ResDOC * DOC - DOSed - 4.57 * Nitr$$

Appendix B: Model Parameters

Symbols	Description	Units	Values
Abso	Adsorption coefficients for PO4 and DIC	NA	0.2
alpha	Intercept of linear radiation dependence (for Duckweed)	1/(W/m²)	0.15
ArrhBM	Temperature-dependent coefficient for phytoplankton basal metabolism	NA	1.02
Arrhdent	Temperature-dependent coefficient for denitrification	NA	1.08
ArrhDO	Temperature-dependent coefficient for sediment oxygen demand	NA	1.05
ArrhFluxDIC	Temperature-dependent coefficient for DIC sediment flux	NA	1.05
ArrhFluxDOC	Temperature-dependent coefficient for DOC sediment flux	NA	1.05
ArrhFluxNH4	Temperature-dependent coefficient for NH4 sediment flux	NA	1.05
ArrhFluxNO3	Temperature-dependent coefficient for NO3 sediment flux	NA	1.05
ArrhFluxPO4	Temperature-dependent coefficient for PO4 sediment flux	NA	1.15
ArrhGr	Temperature-dependent coefficient for phytoplankton growth	NA	1.001
ArrhHyd	Temperature-dependent coefficient for POC hydrolysis	NA	1.08
Arrhloss	Temperature-dependent coefficient for Duckweed loss	NA	1.01
ArrhMinDOC	Temperature-dependent coefficient for DOC mineralisation	NA	1.001
ArrhMinON	Temperature-dependent coefficient for ON mineralisation	NA	1.01
ArrhNit	Temperature-dependent coefficient for nitrification	NA	1.03
ArrhMinOP	Temperature-dependent coefficient for OP mineralisation	NA	1.01
beta	Slope of linear radiation dependence (for Duckweed)	W/m²	0.03
bm	Basal metabolism rate for phytoplankton	1/day	0.06
BM	Algal metabolic rate	1/day	NA
BMDIC	Fraction exuded from phytoplankton as DIC	NA	0.25
BMDOC	Fraction exuded from phytoplankton as DOC	NA	0.5
BMON	Fraction of nitrogen excreted by phytoplankton as ON	NA	0.5
BMOP	Fraction of nitrogen excreted by phytoplankton as OP	NA	0.5

Appendix B: Model Parameters

BMNH4	Fraction of basal metabolism excreted by phytoplankton as NH4	NA	0.5
BMPO4	Fraction of basal metabolism excreted by phytoplankton as PO4	NA	0.5
BMPOC	Fraction of basal metabolism excreted by phytoplankton as POC	NA	0.25
CCDW	Carbon to dry weight ratio for duckweed	mg C/mg dw	0.3
CO2water	CO2 concentration in water column	mg/l	NA
CPhytoplankton	Algal Carbon Biomass	mgC/l	NA
denit	Denitrification rate	1/day	0.06
DIC	Dissolved inorganic carbon	mgC/l	NA
DICsf	Sediment- water flux process for DIC	1/day	NA
DO	Dissolved oxygen	mg O2/l	NA
DOC	Dissolved organic carbon	mgC/l	NA
DOCsf	Sediment-water flux process for DOC	1/day	NA
ON	Organic nitrogen	mgN/l	NA
OP	Organic phosphorus	mgP/l	NA
Dosat	Saturated dissolved oxygen concentration	mg/l	NA
DOsatdoc	DO half saturation constant for mineralization for DOC	mg O2/l	1.5
DOsatdon	DO half saturation constant for mineralization for DON	mg O2/l	0.5
DOsatdop	DO half saturation constant for mineralization for DOP	mg O2/l	0.5
DOSed	Dissolved oxygen sediment demand	1/day	NA
Dweed	Duckweed Biomass	g dw/m ²	NA
FDOM	Fraction of dissolved ON, OP	NA	0.5
Fe	Iron concentration	mg/l	NA
Gmax	Maximum growth rate for phytoplankton	1/day	1.2
GmaxDWeed	Maximum growth rate for duckweed	1/day	0.41
Gr	Phytoplankton growth rate	1/day	NA

Appendix B: Model Parameters

GrDWeed	Duckweed growth rate	1/day	NA
H	water depth	m	NA
H0	Carbonate species	NA	NA
HYD	POC Hydrolysis rate	1/day	NA
hydpoc	Hydrolysis rate for POC	1/day	0.02
Iop	Optimal light intensity for algal growth	W/m ²	100
Isat	Radiation intensity at which growth is saturated	W/m ²	173.6
K	Water-Atmosphere exchange rate for DIC	1/day	0.00018
k	ligh extinction coefficient for phytoplankton	m ⁻¹	NA
KBiomass	limiting effect for duckweed biomass	gC/m ²	26
K0	Solubility of CO2 in water	mgC/l*µatm*day	0.000444
k0	Background extinction of water	m ⁻¹	0.01
k1	Specific ligh extinction coefficient related to phytoplankton	1/m*(mgC/l)	0.02
k2	Specific ligh extinction coefficient related to POC	1/m*(mgC/l)	0.02
k3	Specific ligh extinction coefficient related to DOC	1/m*(mgC/l)	0.01
k4	Specific ligh extinction coefficient related to Duckweed	1/m*(g dw/m ²)	0.16
Kc1	Dissociation constant of H2CO3 at 0 salinity and 20 °C	NA	4.3*(10 [^] (-7))
Kc2	Dissociation constant of HCO3NA at 0 salinity and 20 °C	NA	5.61*(10 [^] (-11))
KhnitDo	Half saturation concentration of DO required for nitrification	mg O2/l	2
Kn	Half saturation constant for nitrogen uptake by phytoplankton	mg N/l	0.065
KnDWeed	Half saturation constant for nitrogen uptake by duckweed	g N/m ²	0.005
Kp	Half saturation constant for phosphorus uptake by phytoplankton	mg P/l	0.006
KpDWeed	Half saturation constant for phosphorus uptake by duckweed	g P/m ²	0.0043
Ligh	Ligh function for Duckweed growth	NA	NA
LimEffBiomass	The limiting effect of Duckweed biomass	NA	NA
loss	Duckweed loss rate	1/day	NA

Appendix B: Model Parameters

lossweed	Loss due to mortality rate of Duckweed	1/day	0.15
Min	DOC mineralisation rate	1/day	NA
mindoc	Mineralization rate for DOC	1/day	0.0057
mindon	Mineralization rate for DON	1/day	0.01
Mindon	DON mineralisation rate	1/day	NA
mindop	Mineralisaton rate for DOP	1/day	0.02
Mindop	DOP mineralisation rate	1/day	NA
NCDW	Nitrogen to dry weight ratio for duckweed	mg N/mg dw	0.03
NCPhy	Nitrogen to Carbon ration	mg N/mg C	0.17
NH4	Ammonium	mg N/l	NA
NH4sf	sediment- water flux process for NH4	1/day	NA
Nitr	Nitrification process	1/day	NA
Nitrmax	Maximum rate for nitrification	1/day	0.1
NO3	Nitrate	mg N/l	NA
NO3sf	Sediment- water flux process for NO3	1/day	NA
PCDW	Phosphorus to dry weight ratio for duckweed	mg P/mg dw	0.0003
pCO2_atm	Partial pressure of CO2 in the atmosphere	µatm	360
PCPhy	Phosphorus to carbon ratio	mg P/mg C	0.024
PH	PH value	NA	8.7
PO4	Phosphorus	mgP/l	NA
POC	Particulate organic carbon	mgC/l	NA
mort	Fraction from duckweed loss term as mortality	NA	0.25
Psf	sediment- water flux process for PO4	1/day	NA
r	nutrients Preference for phytoplankton	NA	NA
Rea	Reaeration rate	1/day	0.1
ResDOC	Dissolved oxygen to carbon ratio in respiration	mg O2 /mg C	2.67

Appendix B: Model Parameters

SedfluxDIC	Sediment–water exchange flux of DIC	1/day	0.02
SedfluxDOC	Sediment–water exchange flux of DOC	1/day	0.025
SedfluxNH4	Sediment–water exchange flux of ammonium	1/day	0.01
SedfluxNO3	Sediment–water exchange flux of NO3	1/day	0.005
SedfluxP	Sediment–water exchange flux of PO4	1/day	0.04
Set	Phytoplankton settling rate	1/day	NA
Setpoc	POC settling rate	1/day	NA
Seton	ON settling rate	1/day	NA
Setop	OP settling rate	1/day	NA
SOD	DO consumption rate on sediment interface	1/day	0.4
TemNit	temperature function for nitrification	NA	NA
Tmin	Minimum temperature for growth fo Duckweed	°C	5
Top	Optimal temperature for phytoplankton growth	°C	26
Uligh	Ligth function for algal growth	NA	NA
UTem	Temperature function for algal growth	NA	NA
UTemDWeed	Temperature function for duckweed growth	NA	NA
Vcph	Settling velocity rate for phytoplankton	m/day	0,02
Vpoc	Settling velocity rate for POC	m/day	0.6
Von	Settling velocity rate for ON	m/day	0.3
Vop	Settling velocity rate for OP	m/day	0.2
Vset	Settling Velocity coefficient for PO4-Fe and PO4-Mn	m/day	0.3